

OIS-010 — Federation Engine

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-009
 - OAS-009 — Federation Architecture
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities of the Federation Engine.

The Constitution defines the principles of Federation.

The Architecture defines the organization of Federation.

This specification defines the executable engine responsible for enabling constitutional cooperation among independent Places.

The Federation Engine SHALL coordinate.

It SHALL never centralize.

Implementation SHALL preserve sovereignty before synchronization.

1. Implementation Objective

The objective of the Federation Engine is to enable independent constitutional Places to exchange understanding while preserving local constitutional authority.

The Engine SHALL preserve:

Sovereignty.

Canonical identity.

Historical continuity.

Relationship integrity.

Independent evolution.

Federation SHALL strengthen cooperation.

It SHALL not weaken autonomy.

2. Federation Lifecycle

Every federated interaction SHALL support the following operational lifecycle.

Discover

↓

Negotiate

↓

Establish Trust

↓

Exchange Capabilities

↓

Exchange Constitutional Objects

↓

Verify Integrity

↓

Integrate Locally

↓

Continue Independent Evolution

Every lifecycle stage SHALL remain observable.

3. Federation Context

The Engine SHALL maintain federation context including:

Remote Place identity.

Supported constitutional versions.

Supported architectural versions.

Schema compatibility.

Trust relationships.

Synchronization history.

Context SHALL remain reconstructable.

4. Capability Negotiation

Before exchanging constitutional objects,

implementations SHALL negotiate operational capabilities.

Capabilities MAY include:

Supported specifications.

Schema versions.

Exchange protocols.

Compression methods.

Authentication methods.

Negotiation SHALL never redefine constitutional behavior.

5. Constitutional Exchange

The Federation Engine SHALL exchange constitutional constructs such as:

Events.

Schemas.

Voices.

Libraries.

Conversations.

Knowledge relationships.

Decisions.

Future constitutional constructs SHALL integrate without redesigning the Engine.

6. Conflict Handling

Federation SHALL preserve constitutional independence.

Conflicting constitutional understanding SHALL remain:

Visible.

Attributable.

Historically preserved.

Implementation SHALL avoid automatic convergence unless constitutionally justified.

7. Synchronization

Synchronization SHALL remain operational.

Synchronization SHALL NOT become constitutional authority.

Temporary divergence SHALL remain constitutionally acceptable.

Equivalent understanding SHALL take precedence over identical storage.

8. Failure Recovery

Federation failures SHALL preserve:

Local sovereignty.

Historical continuity.

Exchange history.

Recoverability.

Implementation SHALL support independent recovery without requiring global coordination.

9. Verification

Verification SHALL demonstrate:

Identity preservation.

Historical reconstruction.

Exchange integrity.

Schema compatibility.

Behavioral equivalence.

Independent constitutional evolution.

10. Federation Pattern

The Federation Engine SHALL optimize interoperability.

It SHALL NOT optimize away sovereignty, diversity, or independent evolution.

Operational convenience SHALL remain subordinate to constitutional integrity.

11. Federation Engine Test

A Federation Engine satisfies this specification when independent implementations, using different infrastructures, communication protocols, storage technologies, and execution models, continuously exchange constitutional understanding while preserving local sovereignty, historical continuity, and architectural compatibility.

Conformance

An implementation conforms to this specification when Federation remains operationally executable, constitutionally sovereign, historically continuous, architecturally compatible, and implementation-independent.

Implementation Principle

Traditional distributed systems synchronize data.

Organodynamics federates understanding.

The Federation Engine therefore exists not to make every Place identical,

but to make every Place capable of learning from every other Place

without surrendering its own history,

its own identity,

or its own constitutional freedom.

End of Specification OIS-010